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Exhaust aftertreatment assembly with a mixer having a mixing plate that is crescent shaped

Abstract

An exhaust aftertreatment assembly includes a tubular conduit and a mixer. The tubular conduit has a central axis. The mixer is disposed in the tubular conduit. The mixer includes a first mixing plate and a second mixing plate. The first mixing plate is crescent shaped. The first mixing plate includes a first plate convex edge and a first plate concave edge. The first plate convex edge is attached to the tubular conduit. The first plate concave edge intersects the first plate convex edge at a first plate first point and a first plate second point. The second mixing plate is crescent shaped. The second mixing plate includes a second plate convex edge and a second plate concave edge. The second plate convex edge is attached to the tubular conduit. The second plate concave edge intersects the second plate convex edge.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) The present application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/545,845, filed on Oct. 26, 2023, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to mixers for exhaust aftertreatment systems for an internal combustion engine.

BACKGROUND

(2) The exhaust of internal combustion engines, such as diesel engines, includes nitrogen oxide (NOx) compounds. It is desirable to reduce NOx emissions to comply with environmental regulations, for example. To reduce NOx emissions, a treatment fluid may be dosed into the exhaust by a doser assembly within an aftertreatment system. The treatment fluid facilitates conversion of a portion of the exhaust into non-NOx emissions, such as nitrogen (N.sub.2), carbon dioxide (CO.sub.2), and water (H.sub.2O), thereby reducing NOx emissions. These aftertreatment systems may include a mixer that facilitates mixing of the treatment fluid and the exhaust. Increased mixing of the treatment fluid and the exhaust may lead to more efficient conversion of NOx to non-NOx emissions. Mixers can take various forms, each of which has benefits and consequences to operation of an engine system. For example, mixers may increase backpressure on an engine, which may decrease power and/or efficiency of an engine system.

SUMMARY

(3) In one embodiment, an exhaust aftertreatment assembly includes a tubular conduit and a mixer. The tubular conduit has a central axis. The mixer is disposed in the tubular conduit. The mixer includes a first mixing plate and a second mixing plate. The first mixing plate is crescent shaped. The first mixing plate includes a first plate convex edge and a first plate concave edge. The first plate convex edge is attached to the tubular conduit. The first plate concave edge intersects the first plate convex edge at a first plate first point and a first plate second point. The second mixing plate is crescent shaped. The second mixing plate includes a second plate convex edge and a second plate concave edge. The second plate convex edge is attached to the tubular conduit. The second plate concave edge intersects the second plate convex edge at a second plate first point and a second plate second point. A plane in which the first mixing plate extends and a plane in which the second mixing plate extends are oblique to the central axis of the tubular conduit. The plane in which the first mixing plate extends intersects the plane in which the second mixing plate extends inside of the tubular conduit.

(4) In another embodiment, an exhaust aftertreatment assembly includes a tubular conduit and a mixer. The tubular conduit has a central axis. The mixer is disposed in the tubular conduit. The mixer includes a first mixing plate and a second mixing plate. The first mixing plate is crescent shaped. The first mixing plate includes a first plate convex edge and a first plate concave edge. The first plate convex edge is attached to the tubular conduit. The second mixing plate is crescent shaped. The second mixing plate includes a second plate convex edge and a second plate concave edge. The second plate convex edge is attached to the tubular conduit. The first mixing plate and the second mixing plate are positioned relative to the tubular conduit such that the central axis extends between the first plate concave edge and the second plate concave edge. A plane in which the first mixing plate extends and a plane in which the second mixing plate extends are oblique to the central axis of the tubular conduit.

(5) In another embodiment, an exhaust aftertreatment assembly includes a tubular conduit having a central axis a mixer disposed in the tubular conduit. The mixer includes a first mixing plate and a second mixing plate. The first mixing plate includes a first plate convex edge attached to the tubular conduit and a first plate concave edge opposite the first plate convex edge. The second mixing plate includes a second plate convex edge attached to the tubular conduit, and a second plate concave edge opposite the second plate convex edge. The first mixing plate and the second mixing plate are positioned relative to the tubular conduit such that the central axis extends between the first plate concave edge and the second plate concave edge. A plane in which the first

mixing plate extends intersects a plane in which the second mixing plate extends inside of the tubular conduit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying Figures, wherein like reference numerals refer to like elements unless otherwise indicated, in which:

(2) FIG. **1** is a block schematic diagram of an example exhaust aftertreatment system;

(3) FIG. **2** is a top view of a portion of the exhaust aftertreatment system of FIG. **1**;

(4) FIG. **3** is a side view of a portion of the exhaust aftertreatment system of FIG. **2**;

(5) FIG. **4** is a cross-sectional view of the exhaust aftertreatment system of FIG. **3** taken along plane A-A in FIG. **2**, according to various embodiments;

(6) FIG. **5** is a view of DETAIL A in FIG. **4**;

(7) FIG. **6** is another view of DETAIL A in FIG. **4**;

(8) FIG. **7** is a cross-sectional view of the exhaust aftertreatment system of FIG. **5** taken along plane B-B in FIG. **5**;

(9) FIG. **8** is a perspective view of the exhaust aftertreatment system of FIG. **3** with a portion of a tubular conduit hidden, according to various embodiments;

(10) FIG. **9** is a cross-sectional view of the exhaust aftertreatment system of FIG. **8** taken along plane A-A in FIG. **2**;

(11) FIG. **10** is a front view of a first mixing plate, according to various embodiments;

(12) FIG. **11** is a front view of a first mixing plate, according to various embodiments;

(13) FIG. **12** is a front view of a second mixing plate, according to various embodiments;

(14) FIG. **13** is a front view of a first mixing plate, according to various embodiments;

(15) FIG. **14** is a front view of a first mixing plate, according to various embodiments;

(16) FIG. **15** is a front view of a first mixing plate, according to various embodiments;

(17) FIG. **16** is a front view of a first mixing plate, according to various embodiments;

(18) FIG. **17** is a front view of a first mixing plate, according to various embodiments;

(19) FIG. **18** is a perspective view of a first mixing plate, according to various embodiments;

(20) FIG. **19** is a front view of a first mixing plate, according to various embodiments;

(21) FIG. **20** is a front view of a first mixing plate, according to various embodiments;

(22) FIG. **21** is a front view of a first mixing plate, according to various embodiments;

(23) FIG. **22** is a perspective view of a first mixing plate, according to various embodiments;

(24) FIG. **23** is a perspective view of a first mixing plate, according to various embodiments;

(25) FIG. **24** is a perspective view of the exhaust aftertreatment system of FIG. **3** with a portion of a tubular conduit hidden, according to various embodiments;

(26) FIG. **25** is another perspective view of the exhaust aftertreatment system of FIG. **3** with a portion of a tubular conduit hidden, according to various embodiments;

(27) FIG. **26** is a perspective view of the exhaust aftertreatment system of FIG. **3** with a portion of a tubular conduit hidden, according to various embodiments;

(28) FIG. **27** is a perspective view of a first mixing plate, according to various embodiments;

(29) FIG. **28** is a perspective view of a first mixing plate, according to various embodiments;

(30) FIG. **29** is a perspective view of a first mixing plate, according to various embodiments;

(31) FIG. **30** is a perspective view of a first mixing plate, according to various embodiments;

(32) FIG. **31** is a perspective view of a first mixing plate, according to various embodiments;

(33) FIG. **32** is a perspective view of a first mixing plate, according to various embodiments;

(34) FIG. **33** is a perspective view of a first mixing plate, according to various embodiments;

(35) FIG. **34** is a perspective view of a first mixing plate, according to various embodiments;
(36) FIG. **35** is a perspective view of a first mixing plate, according to various embodiments;
(37) FIG. **36** is a perspective view of the exhaust aftertreatment system of FIG. **3** with a portion of a tubular conduit hidden, according to various embodiments;
(38) FIG. **37** is a perspective view of the exhaust aftertreatment system of FIG. **3** with a portion of a tubular conduit hidden, according to various embodiments;
(39) FIG. **38** is an end view of the exhaust aftertreatment system of FIG. **3**, according to various embodiments;
(40) FIG. **39** is a perspective view of the exhaust aftertreatment system of FIG. **3** with a portion of a tubular conduit hidden, according to various embodiments;
(41) FIG. **40** is an end view of the exhaust aftertreatment system of FIG. **3**, according to various embodiments;
(42) FIG. **41** is an end view of the exhaust aftertreatment system of FIG. **3**, according to various embodiments;
(43) FIG. **42** is an end view of the exhaust aftertreatment system of FIG. **3**, according to various embodiments;
(44) FIG. **43** is an end view of the exhaust aftertreatment system of FIG. **3**, according to various embodiments;
(45) FIG. **44** is an end view of the exhaust aftertreatment system of FIG. **3**, according to various embodiments;
(46) FIG. **45** is an end view of the exhaust aftertreatment system of FIG. **3**, according to various embodiments;
(47) FIG. **46** is an end view of the exhaust aftertreatment system of FIG. **3**, according to various embodiments;
(48) FIG. **47** is an end view of the exhaust aftertreatment system of FIG. **3**, according to various embodiments;
(49) FIG. **48** is a perspective view of a first mixing plate, according to various embodiments;
(50) FIG. **49** is a perspective view of a first mixing plate, according to various embodiments;
(51) FIG. **50** is a perspective view of a first mixing plate, according to various embodiments;
(52) FIG. **51** is a perspective view of a first mixing plate, according to various embodiments;
(53) FIG. **52** is a perspective view of a first mixing plate, according to various embodiments;
(54) FIG. **53** is a perspective view of a first mixing plate, according to various embodiments;
(55) FIG. **54** is an end view of the exhaust aftertreatment system of FIG. **3**, according to various embodiments;
(56) FIG. **55** is a perspective view of a first mixing plate, according to various embodiments; and
(57) FIG. **56** is a perspective view of a first mixing plate, according to various embodiments.
(58) It will be recognized that the Figures are schematic representations for purposes of illustration. The Figures are provided for the purpose of illustrating one or more implementations with the explicit understanding that the Figures will not be used to limit the scope or the meaning of the claims.

DETAILED DESCRIPTION

(59) Following below are more detailed descriptions of various concepts related to, and implementations of, methods, apparatuses, and for providing a mixer for an exhaust aftertreatment assembly of an internal combustion engine. The various concepts introduced above and discussed in greater detail below may be implemented in any of a number of ways, as the described concepts are not limited to any particular manner of implementation. Examples of specific implementations and applications are provided primarily for illustrative purposes.

I. Overview

(60) Internal combustion engines (e.g., diesel internal combustion engines, etc.) produce exhaust that is often treated by a doser assembly within an exhaust aftertreatment system. The doser

assembly typically treats exhaust using a treatment fluid (e.g., reductant, hydrocarbon, etc.) released from the doser assembly by an injector of a doser. The treatment fluid, such as the reductant, may be adsorbed by a catalyst member. The adsorbed treatment fluid in the catalyst member functions to reduce NOx in the exhaust. The treatment fluid, such as the hydrocarbon, may increase a temperature of the exhaust to reduce NOx in the exhaust. The doser assembly is mounted on a component of the exhaust aftertreatment system. For example, the doser assembly may be mounted on a decomposition reactor, an exhaust conduit, a panel, or other similar components of the exhaust aftertreatment system.

(61) Mixing the exhaust with the treatment fluid improves the reduction of NOx in the exhaust. A device can be used to facilitate mixing between the exhaust and the treatment fluid through turbulent flow (e.g., turbulence, etc.). Turbulence in the form of swirling (e.g., eddies, etc.) improves the mixing characteristics of a fluid. For example, swirling of the exhaust causes dispersal of treatment fluid within the exhaust, thereby improving the mixing between the exhaust and the treatment fluid. However, a device in a flow path of the treatment fluid may be prone to collecting (e.g., accumulating, etc.) deposits of the treatment fluid. These deposits may reduce a mixing efficiency of the device and a flow rate of the exhaust and/or the treatment fluid within a conduit that the device is within or fluidly coupled to.

(62) Implementations herein are directed to an exhaust aftertreatment system that includes a tubular conduit that is configured to receive exhaust and treatment fluid with a mixer disposed inside of the tubular conduit. The mixer includes crescent shaped mixing plates which each extend on a plane that is oblique to the central axis of the tubular conduit. The angle of the crescent shaped mixing plates relative to the central axis facilitate turbulent, spiraling flow of the exhaust and the treatment fluid through the mixer body by directing exhaust and treatment fluid towards an outer periphery of the tubular conduit. The swirling motion also induces shear on downstream faces of the crescent shaped mixing blades. As a result of this shear, formation of deposits of the treatment fluid on the downstream faces of the crescent shaped mixing blades is prevented or minimized. The mixing plates may be arranged in the tubular conduit such that the planes in which each mixing plate extend intersect inside of the tubular conduit. The mixing plates may be flat and may be positioned such that the central axis extends in between the concave edges of the mixing plates. These arrangements may beneficially provide mixing of the exhaust and treatment fluid while minimizing cost relative to other mixers that are suspended within a conduit, for example.

II. Overview of Exhaust Aftertreatment Systems

(63) FIG. **1** depicts an exhaust aftertreatment system **100** having an example treatment fluid delivery system **102** for an exhaust conduit system **104**. The exhaust aftertreatment system **100** includes the treatment fluid delivery system **102**, a particulate filter **106** (e.g., a diesel particulate filter (DPF)), a tubular conduit **108**, and a catalyst member **110** (e.g., SCR catalyst member, etc.).

(64) The exhaust aftertreatment system **100** includes an exhaust aftertreatment assembly **111**. The exhaust aftertreatment assembly **111** includes the tubular conduit **108**. In various embodiments, the exhaust aftertreatment assembly **111** includes other components of the exhaust aftertreatment system **100**, such as components of the treatment fluid delivery system **102**.

(65) The particulate filter **106** is configured to (e.g., structured to, able to, etc.) remove particulate matter, such as soot, from exhaust flowing in the exhaust conduit system **104**. The particulate filter **106** includes an inlet, where the exhaust is received, and an outlet, where the exhaust exits after having particulate matter substantially filtered from the exhaust and/or converting the particulate matter into carbon dioxide. In some implementations, the particulate filter **106** may be omitted.

(66) The tubular conduit **108** functions as a decomposition chamber (e.g., decomposition chamber, reactor, reactor pipe, conduit, etc.). The tubular conduit **108** is configured to receive the exhaust from the particulate filter **106** and a treatment fluid from the treatment fluid delivery system **102**. The treatment fluid may be, for example, a reductant (e.g., urea, diesel exhaust fluid (DEF), Adblue®, a urea water solution (UWS), an aqueous urea solution (e.g., AUS32, etc.), and/or other

similar fluids) or a hydrocarbon (e.g., fuel, oil, additive, etc.). When the reductant is introduced into the exhaust, reduction of emission of undesirable components (e.g., NOx, etc.) in the exhaust may be facilitated. When the hydrocarbon is introduced into the exhaust, the temperature of the exhaust may be increased (e.g., to facilitate regeneration of components of the exhaust aftertreatment system **100**, etc.). For example, the exhaust aftertreatment system **100** may include a spark plug **109** (e.g., igniter, etc.) configured to increase the temperature of the exhaust by combusting the hydrocarbon within the exhaust. The tubular conduit **108** includes an inlet in fluid communication with the particulate filter **106** to receive the exhaust containing NOx emissions and an outlet for the exhaust, NOx emissions, ammonia, and/or treatment fluid to flow to the catalyst member **110**.

(67) The treatment fluid delivery system **102** includes a doser assembly **112** (e.g., dosing module, etc.) configured to dose the treatment fluid into the tubular conduit **108** (e.g., via an injector). The doser assembly **112** is mounted to the tubular conduit **108** such that the doser assembly **112** may dose the treatment fluid into the exhaust flowing through the exhaust conduit system **104**. The doser assembly **112** may include an insulator (e.g., vibrational insulator, thermal insulator, etc.) interposed between a portion of the doser assembly **112** and a portion of the tubular conduit **108** on which the doser assembly **112** is mounted. The insulator may mitigate transfer of vibrations and/or heat from the tubular conduit **108** to the doser assembly **112**.

(68) The doser assembly **112** is fluidly coupled to (e.g., fluidly configured to communicate with, etc.) a treatment fluid source **114**. The treatment fluid source **114** may include multiple treatment fluid sources **114**. The treatment fluid source **114** may be, for example, a diesel exhaust fluid tank containing Adblue®. A treatment fluid pump **116** (e.g., supply unit, etc.) is used to pressurize the treatment fluid from the treatment fluid source **114** for delivery to the doser assembly **112**. In some embodiments, the treatment fluid pump **116** is pressure-controlled (e.g., controlled to obtain a target pressure, etc.). The treatment fluid pump **116** includes a treatment fluid filter **118**. The treatment fluid filter **118** filters (e.g., strains, etc.) the treatment fluid prior to the treatment fluid being provided to internal components (e.g., pistons, vanes, etc.) of the treatment fluid pump **116**. For example, the treatment fluid filter **118** may inhibit or prevent the transmission of solids (e.g., solidified treatment fluid, contaminants, etc.) to the internal components of the treatment fluid pump **116**. In this way, the treatment fluid filter **118** may facilitate prolonged desirable operation of the treatment fluid pump **116**. In some embodiments, the treatment fluid pump **116** is coupled (e.g., fastened, attached, affixed, welded, etc.) to a chassis of a vehicle associated with the exhaust aftertreatment system **100**.

(69) The doser assembly **112** includes at least one injector **120**. Each injector **120** is configured to dose the treatment fluid into the exhaust (e.g., within the tubular conduit **108**, etc.) along an injection axis **119**. The exhaust aftertreatment assembly **111** also includes a mixer **121** (e.g., a swirl generating device, a vane plate, inlet plate, deflector plate, etc.) (e.g., in addition to the tubular conduit **108**, etc.). At least a portion of the mixer **121** is located within the tubular conduit **108**. The mixer **121** is configured to receive exhaust from the tubular conduit **108** and treatment fluid from the injector **120**, such that the injection axis **119** extends into the mixer **121**. The mixer **121** is also configured to facilitate mixing of the exhaust and the treatment fluid. The mixer **121** is configured to facilitate swirling (e.g., tumbling, rotation, etc.) of the exhaust and mixing (e.g., combination, etc.) of the exhaust and the treatment fluid so as to disperse the treatment fluid within the exhaust downstream of the mixer **121**. By dispersing the treatment fluid within the exhaust (e.g., to obtain an increased uniformity index, etc.) using the mixer **121**, reduction of emission of undesirable components in the exhaust is enhanced or a temperature of the exhaust may be increased.

(70) In some embodiments, the treatment fluid delivery system **102** also includes an air pump **122**. In these embodiments, the air pump **122** draws air from an air source **124** (e.g., air intake, etc.) and through an air filter **126** disposed upstream of the air pump **122**. Additionally, the air pump **122** provides the air to the doser assembly **112** via a conduit. In these embodiments, the doser assembly **112** is configured to mix the air and the treatment fluid into an air-treatment fluid mixture and to

provide the air-treatment fluid mixture into the tubular conduit **108**. In other embodiments, the treatment fluid delivery system **102** does not include the air pump **122** or the air source **124**. In such embodiments, the doser assembly **112** is not configured to mix the treatment fluid with air.

(71) The spark plug **109**, the doser assembly **112**, and the treatment fluid pump **116** are also electrically or communicatively coupled to a treatment fluid delivery system controller **128**. The treatment fluid delivery system controller **128** may control the spark plug **109** to ignite the treatment fluid in the tubular conduit **108**. The treatment fluid delivery system controller **128** controls the doser assembly **112** to dose the treatment fluid into the tubular conduit **108**. The treatment fluid delivery system controller **128** may also control the treatment fluid pump **116**.

(72) The treatment fluid delivery system controller **128** includes a processing circuit **130**. The processing circuit **130** includes a processor **132** and a memory **134**. The processor **132** may include a microprocessor, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), etc., or combinations thereof. The memory **134** may include, but is not limited to, electronic, optical, magnetic, or any other storage or transmission device capable of providing a processor, ASIC, FPGA, etc. with program instructions. This memory **134** may include a memory chip, Electrically Erasable Programmable Read-Only Memory (EEPROM), Erasable Programmable Read Only Memory (EPROM), flash memory, or any other suitable memory from which the treatment fluid delivery system controller **128** can read instructions. The instructions may include code from any suitable programming language. The memory **134** may include various modules that include instructions which are configured to be implemented by the processor **132**.

(73) In various embodiments, the treatment fluid delivery system controller **128** is configured to communicate with a central controller **136** (e.g., engine control unit (ECU), engine control module (ECM), etc.) of an internal combustion engine having the exhaust aftertreatment system **100**. In some embodiments, the central controller **136** and the treatment fluid delivery system controller **128** are integrated into a single controller.

(74) In some embodiments, the central controller **136** is communicable with a display device (e.g., screen, monitor, touch screen, heads up display (HUD), indicator light, etc.). The display device may be configured to change state in response to receiving information from the central controller **136**. For example, the display device may be configured to change between a static state (e.g., displaying a green light, displaying a “SYSTEM OK” message, etc.) and an alarm state (e.g., displaying a blinking red light, displaying a “SERVICE NEEDED” message, etc.) based on a communication from the central controller **136**. By changing state, the display device may provide an indication to a user (e.g., operator, etc.) of a status (e.g., operation, in need of service, etc.) of the treatment fluid delivery system **102**.

(75) The tubular conduit **108** is located upstream of the catalyst member **110**. As a result, the treatment fluid is injected upstream of the catalyst member **110** such that the catalyst member **110** receives a mixture of the treatment fluid and exhaust. The treatment fluid droplets undergo the processes of evaporation, thermolysis, and hydrolysis to form non-NOx emissions (e.g., gaseous ammonia, etc.) within the exhaust conduit system **104**.

(76) The catalyst member **110** includes an inlet in fluid communication with the tubular conduit **108** from which exhaust and treatment fluid are received and an outlet in fluid communication with an end of the exhaust conduit system **104**.

(77) The exhaust aftertreatment system **100** may further include an oxidation catalyst member (e.g., a diesel oxidation catalyst (DOC)) in fluid communication with the exhaust conduit system **104** (e.g., downstream of the catalyst member **110** or upstream of the particulate filter **106**) to oxidize hydrocarbons and carbon monoxide in the exhaust.

(78) In some implementations, the particulate filter **106** may be positioned downstream of the tubular conduit **108**. For instance, the particulate filter **106** and the catalyst member **110** may be combined into a single unit. In some implementations, the doser assembly **112** may instead be positioned downstream of a turbocharger or upstream of a turbocharger.

(79) The exhaust aftertreatment system **100** also includes a doser mounting bracket **138** (e.g., mounting bracket, coupler, plate, etc.). The doser mounting bracket **138** couples the doser assembly **112** to a component of the exhaust aftertreatment system **100**. The doser mounting bracket **138** is configured to mitigate the transfer of heat from the exhaust passing through the exhaust conduit system **104** to the doser assembly **112**. In this way, the doser assembly **112** is capable of operating more efficiently and desirably than other doser assemblies which are not able to mitigate the transfer of heat. Additionally, the doser mounting bracket **138** is configured to aid in reliable installation of the doser assembly **112**. This may decrease manufacturing costs associated with the exhaust aftertreatment system **100** and ensure repeated desirable installation of the doser assembly **112**.

(80) In various embodiments, the doser mounting bracket **138** couples the doser assembly **112** to the tubular conduit **108**. In some embodiments, the doser mounting bracket **138** couples the doser assembly **112** to an exhaust conduit of the exhaust conduit system **104**. For example, the doser mounting bracket **138** may couple the doser assembly **112** to an exhaust conduit of the exhaust conduit system **104** that is upstream of the tubular conduit **108** or to an exhaust conduit of the exhaust conduit system **104** that is downstream of the tubular conduit **108**. In some embodiments, the doser mounting bracket **138** couples the doser assembly **112** to the particulate filter **106** and/or the catalyst member **110**. The location of the doser mounting bracket **138** may be varied depending on the application of the exhaust aftertreatment system **100**. For example, in some exhaust aftertreatment systems **100**, the doser mounting bracket **138** may be located further upstream than in other exhaust aftertreatment systems **100**. Furthermore, some exhaust aftertreatment systems **100** may include multiple doser assemblies **112** and therefore may include multiple doser mounting brackets **138**.

(81) FIGS. **2-9** illustrate various embodiments of the exhaust aftertreatment system **100**. The doser assembly **112** is configured to inject treatment fluid into the tubular conduit **108**. The injection axis **119** may extend into the tubular conduit **108** at an angle relative to a central axis **205** of the tubular conduit **108**. For example, in some embodiments, the injection axis **119** may be coincident with the central axis **205** of the tubular conduit **108**. In other embodiments, the injection axis **119** may be perpendicular to the central axis **205** of the tubular conduit **108**. In yet other embodiment, the injection axis **119** may be parallel to the central axis **205** of the tubular conduit **108**.

III. Overview of Example Mixers

(82) FIGS. **4-9** illustrate portions of the exhaust aftertreatment system **100** including the mixer **121**, according to various embodiments. The mixer **121** includes crescent shaped mixing blades, including a first mixing plate **200** (e.g., blade, etc.) and a second mixing plate **201** (e.g., blade, etc.). As is explained in more detail herein, the first mixing plate **200** and the second mixing plate **201** facilitate swirling of the exhaust within the tubular conduit **108**, and therefore swirling and mixing of the exhaust and the treatment fluid. FIG. **10** illustrates a flat view of the first mixing plate **200** according to various embodiments. The first mixing plate **200** may include an edge chamfer to provide a flush fit between the first mixing plate **200** and the tubular conduit **108**.

(83) In various embodiments, the second mixing plate **201** is identical to the first mixing plate **200**. It is understood that one configuration for the first mixing plate **200** may be utilized for the first mixing plate **200** when the same or another configuration for the first mixing plate **200** may be utilized for the second mixing plate **201**. Similarly, description herein of the first mixing plate **200** similarly applies to any other mixing plates (e.g., the second mixing plate **201**, etc.) unless otherwise indicated to the contrary.

(84) As utilized herein, “crescent shaped” means a shape that is bounded by two circular arcs of unequal radii, where the shape does not include the center of either circle defining the circular arcs. A “crescent” is a type of lune which is a shape that is bounded by two circular arcs of unequal radii.

(85) In some embodiments, the second mixing plate **201** is identical to the first mixing plate **200**. This identical relationship may reduce complexity associated with manufacturing the exhaust

aftertreatment system **100**. In other embodiments, the second mixing plate **201** is different from the first mixing plate **200**. These differences may enable tailoring of the exhaust aftertreatment system **100** to produce a target effect on the flow of the exhaust. This target effect may be causing the treatment fluid to have a target uniformity index within the exhaust, for example. In some embodiments, the exhaust aftertreatment system **100** includes only a single mixing plate, the first mixing plate **200**, and does not include the second mixing plate **201**.

(86) The tubular conduit **108** is configured to receive treatment fluid and exhaust through an inlet end **202** of the tubular conduit. The mixer **121** induces an at least partially rotational flow of a mixture of the exhaust and treatment fluid. The mixer **121** disrupts and directs the flow of exhaust around the central axis **205** to cause rotation of the exhaust and the treatment fluid within the tubular conduit **108** to facilitate mixing of the exhaust and the treatment fluid within the tubular conduit **108** and downstream of the tubular conduit **108**.

(87) As shown in FIG. **4**, the mixer **121** also includes a fine mixer **203** (e.g., in addition to the first mixing plate **200** and the second mixing plate **201**, etc.). The fine mixer **203** includes a perforated plate with one or more angled projections (e.g., baffles, etc.) such that exhaust may flow directly through the fine mixer and into a region between the first mixing plate **200** and second mixing plate **201**. The fine mixer **203** is located downstream of the doser assembly **112** and upstream of the first mixing plate **200**. The fine mixer **203** disrupts flow and increases turbulence such that the exhaust and treatment fluid begins to mix before the exhaust enters the region between the first mixing plate **200** and second mixing plate **201**. Additionally, the fine mixer **203** functions to break up droplets of the treatment fluid received from the doser assembly **112**.

(88) In various embodiments, the mixer **121** also includes a perforated plate **204**. The perforated plate **204** is located downstream of the second mixing plate **201**, such that if there are more than two mixing plates (e.g., a third mixing plate in addition to the first mixing plate **200** and the second mixing plate **201**, etc.) in the mixer **121**, the perforated plate **204** is located between the most downstream mixing plate and an outlet of the tubular conduit **108**. The perforated plate **204** disrupts flow such that it reduces the turbulence of the flow prior to the flow being provided to the catalyst member **110**. The perforated plate **204** increases the laminar region of the exhaust flow before the exhaust enters the region downstream of the tubular conduit **108**, such as the catalyst member **110**. In this way, the perforated plate **204** enhances operation of the catalyst member **110**.

(89) The fine mixer **203** and/or the perforated plate **204** may be omitted in some applications. For example, the mixer **121** may include only the first mixing plate **200**, the second mixing plate **201**, and the perforated plate **204** downstream of the second mixing plate **201**, and not include the fine mixer **203**. The mixer **121** may also include only the first mixing plate **200**, the second mixing plate **201**, and the fine mixer **203** located upstream of the first mixing plate **200**, and not include the perforated plate **204**. In another example, the mixer **121** may include only the first mixing plate **200** and the second mixing plate **201**, and not include the fine mixer **203** or the perforated plate **204**.

(90) As shown in FIG. **5**, the first mixing plate **200** extends in a plane **5-5** which is oblique to (e.g., angled relative to, not perpendicular with, not parallel to, not perpendicular with and not parallel to, etc.) the central axis **205** of the tubular conduit **108** (e.g., see FIG. **5** angle A1). As shown in FIG. **6**, the second mixing plate **201** also extends in a plane **6-6** which is oblique to the central axis **205** of the tubular conduit **108** (e.g., see FIG. **6** angle A2). The angle of the plane **5-5** in which the first mixing plate **200** and the angle of the plane **6-6** in which the second mixing plate **201** extends relative to the central axis **205** of the tubular conduit **108** (e.g., A1 and A2, respectively) enables rotation of the exhaust and treatment fluid within the tubular conduit **108** by directing exhaust and fluid toward the outer periphery of the tubular conduit **108**.

(91) The angles for all of the mixing plates (e.g., the first mixing plate **200**, the second mixing plate **201**, etc.) are all measured in the same direction (e.g., clockwise, counterclockwise, etc.). As depicted in FIGS. **5** and **9** and described herein, A1 and A2 are measured in the clockwise direction. In other embodiments, A1 and A2 are measured in the counterclockwise direction (i.e., the same

angular amounts described herein are instead measured in the counterclockwise direction). For example, the discussion of A1 being between 181° and 269°, inclusive, below is depicted in the clockwise direction, but it is understood that the A1 may also be between 181° and 269°, inclusive, in the counterclockwise direction.

(92) In various embodiments, A1 is between 181° and 269°, inclusive. In various embodiments, A2 is between 91° and 179°, inclusive. In various embodiments, A1 is between 200° and 250°, inclusive. In various embodiments, A2 is between 110° and 170°, inclusive. Additionally, in some embodiments, A1 is between 200° and 250°, inclusive, and A2 is between 110° and 170°, inclusive. In some embodiments, A1 is between 22° and 230°, inclusive. In some embodiments, A2 is between 130° and 140°, inclusive.

(93) A relationship between A1 and A2 is important to operation of the mixer **121**. In various embodiments, A1 is equal to A2+X, where X is between 90° and 110°. In some embodiments, X is 0100.

(94) As shown in FIG. **9**, in some embodiments, the oblique angle of the plane in which the first mixing plate **200** and the second mixing plate **201** extend (e.g., A1 and A2, respectively) relative to the central axis **205** may be selected such that A2=360°−A1. As a result, the first mixing plate **200** and second mixing plate **201** mirror each other relative to the central axis **205** of the tubular conduit **108**. In various embodiments, A1 is equal to 225° and A2 is equal to 135°.

(95) As illustrated in FIG. **4**, the first mixing plate **200** includes a first plate concave edge **210** and a first plate convex edge **209**. The first plate convex edge **209** intersects the first plate concave edge **210** at a first plate first point **211** and a first plate second point **213** (shown in FIG. **8**). The first plate convex edge **209** of the first mixing plate **200** enables the first mixing plate **200** to be coupled to the tubular conduit **108** along an entire outer periphery of the first mixing plate **200** (e.g., the first plate convex edge **209**). This enhances structural stability and minimizes bypassing of the rotational region between the first mixing plate **200** and second mixing plate **201** by the exhaust traveling between the tubular conduit **108** and the first mixing plate **200**.

(96) As discussed above, every mixer is defined by a backpressure. The mixer **121** is configured such that the backpressure is minimized. One aspect of this configuration is the first plate concave edge **210** which provides a path for exhaust to travel along the central axis **205**. The first plate convex edge **209** also facilitates rotational flow of the exhaust as the exhaust passes over the first mixing plate **200** and into the downstream region of the tubular conduit **108**.

(97) Similarly, the second mixing plate **201** includes a second plate convex edge **215** and a second plate concave edge **216**. The second plate concave edge **216** intersects the second plate convex edge **215** at a second plate first point **217** and a second plate second point **219**. The second plate convex edge **215** of the second mixing plate **201** enables the second mixing plate **201** to be coupled to the tubular conduit **108** along an entire outer periphery of the second mixing plate **201** (e.g., the second plate convex edge **215**). This enhances structural stability and minimizes bypassing of the rotational region between the first mixing plate **200** and second mixing plate **201** by traveling between the tubular conduit **108** and the second plate convex edge **215**.

(98) Another aspect of the configuration of the mixer **121** that provides minimized backpressure is the second plate concave edge **216** which provides a path for exhaust to travel along the central axis **205**. The second plate concave edge **216** also facilitates rotational flow of the exhaust as the exhaust passes over the second mixing plate **201**.

(99) As illustrated in FIG. **9**, the planes in which the first mixing plate **200** and second mixing plate **201** extend intersect in the tubular conduit **108**. In some embodiments, the plane **5-5** in which the first mixing plate **200** extends intersects the second mixing plate **201** at a midpoint of the second plate convex edge **215** (e.g., between the second plate first point **217** and the second plate second point **219**). In other embodiments, the plane **5-5** along which the first mixing plate **200** extends intersects the second mixing plate **201** at another point along the second plate convex edge **215**. In other embodiments, the first mixing plate **200** and the second mixing plate **201** are spaced apart

along the central axis **205** such that the plane **5-5** in which the first mixing plate **200** extends will not intersect the plane **6-6** in which the second mixing plate **201** extends within the tubular conduit **108**.
(100) In some embodiments, such as illustrated in FIG. **5**, the first mixing plate **200** and the second mixing plate **201** are flat (e.g., disposed along a plane, not bent, not twisted). In these embodiments, the first mixing plate **200**, the first plate convex edge **209**, and first plate concave edge **210** are all coincident to the plane **5-5** In which the first mixing plate **200** extends inside of the tubular conduit **108**. The second mixing plate **201**, the second plate convex edge **215**, and second plate concave edge **216** are also all coincident to the plane **6-6** in which the second mixing plate **201** extends inside of the tubular conduit **108**.
(101) In other embodiments, the first mixing plate **200** is not flat, such that the first plate convex edge **209**, the first plate concave edge **210**, and/or all or part of the first mixing plate **200** are not all coincident with the plane **5-5** in which the first mixing plate **200** extends. For example, the first mixing plate **200** may be helically shaped such that the first plate convex edge **209**, the first plate concave edge **210**, and the first mixing plate **200** are not coincident with the same plane **5-5** in which the first mixing plate **200** extends. In other embodiments, the second mixing plate **201** is not flat, such that the second plate convex edge **215**, the second plate concave edge **216**, and/or all or part of the second mixing plate **201** are not all entirely coincident with the plane **6-6** in which the second mixing plate extends. The second mixing plate **201** may be helically shaped such that the second plate convex edge **215**, the second plate concave edge **216**, and the second mixing plate **201** are not coincident with the same plane **6-6** in which the second mixing plate **201** extends.
(102) The first mixing plate **200** and second mixing plate **201** are attached to the tubular conduit **108** such that the central axis **205** of the tubular conduit **108** extends between the first plate convex edge **209** and the second plate concave edge **216** (e.g., the central axis **205** does not intersect the first mixing plate **200** or the second mixing plate **201**). As a result, as illustrated in FIG. **7**, a line segment L could be drawn from the tubular conduit inlet view A-A coincident with the first plate convex edge **209** and second plate convex edge **215** through the central axis **205** of the tubular conduit **108**. This arrangement mitigates backpressure because there is a path through which exhaust can flow directly between the first mixing plate **200** and second mixing plate **201** without being blocked by the first mixing plate **200** or the second mixing plate **201**.
(103) In some embodiments, such as illustrated in FIG. **5** and FIG. **8**, the first mixing plate **200** or the second mixing plate **201**, or both, do not include perforations or textured surfaces. Such a configuration of the first mixing plate **200** and the second mixing plate **201** may decrease a cost associated with producing the first mixing plate **200** and the second mixing plate **201**.
(104) In other embodiments, either the first mixing plate **200** or second mixing plate **201**, or both, include perforations and/or textured surfaces (e.g., louvers, flanges, raised patterns, or circular perforations). Perforations may mitigate backpressure by facilitating another path through which exhaust could travel through the first mixing plate **200** and/or the second mixing plate **201**. The perforations may cover the entirety of the first mixing plate **200** and/or second mixing plate **201**. The textured surfaces may cover the entirety of the first mixing plate **200** and/or second mixing plate **201**. The textured surfaces may cover only one side (e.g., the upstream side) of the first mixing plate **200** and/or second mixing plate **201**.
(105) In some embodiments, either the first mixing plate **200** or second mixing plate **201**, or both, have an airfoil shape. Such a configuration may provide flow along the first mixing plate **200** and/or the second mixing plate **201** in a manner that enhances swirl and therefore mixing of the treatment fluid and exhaust.
(106) In one embodiment, as illustrated in FIG. **8**, the mixer **121** includes a third mixing plate **301** and a fourth mixing plate **302**. The third mixing plate **301** and the fourth mixing plate **302** are positioned at a 90-degree angle measured clockwise around the central axis **205** of the tubular conduit **108** relative to the first mixing plate **200** or the second mixing plate **201**. A midpoint of a

third plate convex edge **500** of the third mixing plate **301** is positioned 90 degrees clockwise relative to a midpoint of the first plate convex edge **209** of the first mixing plate **200** and the central axis **205** of the tubular conduit **108**.

(107) The third mixing plate **301** may be positioned such that a midpoint of the third plate convex edge **500** is positioned 90 degrees measured clockwise around the central axis **205** of the tubular conduit **108** relative to a midpoint of the first plate convex edge **209** or the second plate convex edge **215**. Arranging additional mixing plates (e.g., the third mixing plate **301** and fourth mixing plate **302**) at 90-degree angles facilitates a continuous rotational flow of exhaust between the first mixing plate **200** and most downstream mixing plate (e.g., the fourth mixing plate **302** in FIG. **8**) in the tubular conduit **108**.

(108) FIG. **11** illustrates an embodiment of the first mixing plate **200** being a truncated crescent shape with a first connecting edge **221** and a second connecting edge **224**. The first plate convex edge **209** and first plate concave edge **210** intersect the first connecting edge **221** at a first connecting point **222** and a second connecting point **223**, respectively. The first plate convex edge **209** and first plate concave edge **210** intersect the second connecting edge **224** at a third connecting point **225** and a fourth connecting point **226**, respectively.

(109) FIG. **12** illustrates an embodiment of the second mixing plate **201** being a truncated crescent shape with a third connecting edge **230** and a fourth connecting edge **231**. The second plate convex edge **215** and second plate concave edge **216** intersect the third connecting edge **230** at a fifth connecting point **232** and a sixth connecting point **233**, respectively. The second plate convex edge **215** and second plate concave edge **216** intersect the fourth connecting edge **231** at a third connecting point **234** and a fourth connecting point **2356**, respectively.

(110) The aforementioned connecting edges may be straight (e.g., linear), circular, or an irregular geometric shape (e.g., a line that bends/curves between the distal and proximal end). The mixer **121** may possess a combination of crescent shaped mixing plates which have convex and concave edges that intersect at a point (e.g., as shown in FIG. **4**), and crescent shaped mixing plates with boundaries (e.g., as shown in FIGS. **11** and **12**).

(111) The first mixing plate **200** is formed from a blank **1300** (e.g., stamping, etc.). The blank **1300** is elliptical such that the first plate convex edge **209** extends along the tubular conduit **108**. The blank **1300** cannot be circular because of the angle A1 at which the first mixing plate **200** is located. The blank **1300** is shown in FIGS. **13-19**, according to various embodiments. The tubular conduit **108** has a first radius R1. The first mixing plate **200** has a longest diameter and a shortest diameter. In various embodiments, the shortest diameter of the first mixing plate **200** is equal 2R1.

(112) In FIGS. **13-17**, the first plate concave edge **210** is formed using a cut formed by a single circle with a second radius R2. In various embodiments, R2 is between 0.55R1 and 2R1, inclusive. In some embodiments, R2 is equal to R1.

(113) The single circle is located such that a center of the single circle is separated from a center of the blank **1300** by a first separation S1. S1 is less than R2. Additionally, the first mixing plate **200** is configured such that 0.01(R2−S1) and 0.9(R2−S1), inclusive. In various embodiments, R2−S1 is equal to 0.15T, where T is the longest diameter of the blank **1300**.

(114) The first mixing plate **200** can be tailored for a target application by varying R2 and S1. FIGS. **13-17** illustrate various variations of the first mixing plate **200** with different R2 and S1.

(115) FIG. **19** illustrates an embodiment where the first plate concave edge **210** is formed using a cut formed by a second circle with a second radius R3 and a third circle with a third radius R4. Use of two circles to form the first plate concave edge **210** rather than one circle provides additional capabilities for fine tuning performance of the first mixing plate **200**, such as the impact of the first mixing plate **200** on increasing back pressure and adjusting intensity of swirl provided by the first mixing plate **200**. The first plate concave edge **210** includes two flat portions that extend between the second circle and the first circle adjacent junctions between the second circle and the first circle. These flat portions enable removal of a point that otherwise extends along the junction

between the second circle and the first circle.
(116) The center of the second circle is separated from the center of the blank **1300** by a second separation S2, and the center of the third circle is separated from the center of the second circle by a third separation S3. R3 is less than R4. S2 is less than R3, and S3 is less than R4. Additionally, S2 is less than S3.
(117) The blank **1300** is cut (e.g., using a punch, using a laser, etc.) such that material overlapped by the first circle and material overlapped by the second circle is removed. This removal forms the first plate concave edge **210**, which has three portions, two of which are arcs of the second circle and one of which is an arc of the first circle. The arc of the first circle is positioned between the two arcs of the second circle. The two sections of the first plate concave edge **210** adjoining the arc of the first circle to the arcs of the second circle includes flat portions that avoid creation of points between the arc of the first circle and the arcs of the second circle.
(118) FIGS. **20** and **21** illustrate formation of the first plate concave edge **210** using a cut formed by a non-circular shape (e.g., square, rectangle, ellipse, triangular, etc.). Use of a non-circular shape to form the first plate concave edge **210** rather than one or more circles provides an ability to tune the back pressure and/or intensity of swirl while providing a mechanism for increasing impingement of particles on the first mixing plate **200**.
(119) FIGS. **22** and **23** illustrate the first mixing plate **200** according to various embodiments where the first plate concave edge **210** includes a plurality of edge slots **2200** (e.g., cuts, etc.) and a plurality of edge tabs **2202** (e.g., flanges, baffles, etc.). Each of the edge tabs **2202** is positioned between two of the edge slots **2200**. The edge slots **2200** enable each of the edge tabs **2202** to be bent (e.g., deflected, etc.) away from a body of the first mixing plate **200**.
(120) By variously bending each of the edge tabs **2202**, an impact of the first plate concave edge **210** on the exhaust gas can be tailored for a target application. For example, the amount of tumble provided to the exhaust gas by each of the edge tabs **2202** is related to, and therefore can be controlled by, the amount to which each of the edge tabs **2202** is bent. By increasing the bend, increased tumble to the exhaust gas may be provided. Moreover, the location of these increases can be controlled by bending some of the edge tabs **2202** different amounts. For example, the edge tabs **2202** at the ends of the first plate concave edge **210** may be bent more than the edge tabs **2202** between the ends of the first plate concave edge **210**. In various embodiments, each of the edge tabs **2202** is bent an angle Y away from the body of the first mixing plate **200**, where Y is −90° to 90°, inclusive.
(121) The edge tabs **2202** may also be bent in different directions relative to the body of the first mixing plate **200**. For example, some of the edge tabs **2202** may be bent in a clockwise direction and extend over a first side of the body of the first mixing plate **200** while others of the edge tabs **2202** may be bent in a counterclockwise direction and extend over a second side of the body of the first mixing plate **200**, where the second side is opposite the first side. The shape, size, and arrangement of the edge tabs **2202** may be variously selected such that the first mixing plate **200** is tailored for a target application. Additionally, the edge tabs **2202** can provide a direct impingement device, which may increase decomposition of the treatment fluid and improve efficiency of components of the exhaust aftertreatment system **100**, such as the catalyst member **110**.
(122) Each of the edge slots **2200** is defined by a length W. In various embodiments, W is between 0.02R1 and 0.2 R1, inclusive. Some of the edge slots **2200** may have larger W than others of the edge slots **2200**, thereby enabling one or more of the edge tabs **2202** to be bent a larger amount than others of the edge tabs **2202**.
(123) FIGS. **24** and **25** illustrate the mixer **121** including the third mixing plate **301**, the fourth mixing plate **302**, as well as a fifth mixing plate **2400** and a sixth mixing plate **2402**. The fifth mixing plate **2400** is positioned between the first mixing plate **200** and the second mixing plate **201**, and the sixth mixing plate **2402** is positioned between the third mixing plate **301** and the fourth mixing plate **302**. This creates an overlap between the fifth mixing plate **2400** and both the

first mixing plate **200** and the second mixing plate **201**, as well as an overlap between the sixth mixing plate **2402** and both the third mixing plate **301** and the fourth mixing plate **302**. As discussed above, each of the mixing plates can be variously configured and positioned within the tubular conduit **108** such that the mixer **121** is tailored for a target application.

(124) For example, the mixing plates can be grouped in sets, such as a first set including the first mixing plate **200**, the second mixing plate **201**, and the fifth mixing plate **2400**, and a second set including the third mixing plate **301**, the fourth mixing plate **302**, and the sixth mixing plate **2402**. The angular position of each mixing plate about the central axis **205** of the tubular conduit **108** can be based on to the number of mixing plates in the set divided by 360°. For example, where three mixing plates are included in a set, such as is shown in FIG. **24** with the first mixing plate **200**, the second mixing plate **201**, and the fifth mixing plate **2400**, each mixing plate is separated from another mixing plate by 120°. This results in a complete rotation of the exhaust gas through the set of mixing plates.

(125) FIGS. **26-28** illustrate another embodiment where the first mixing plate **200** and the second mixing plate **201** are profiled to include a bent portion **2600** (e.g., curved portion, etc.) between a first end portion **2602** and a second end portion **2604**. The bent portion **2600** enables the first end portion **2602** and the second end portion **2604** to be positioned closer together or further apart when measured along the central axis **205** of the tubular conduit **108**. Additional bent portions may be included such that a profile of the first mixing plate **200** can be tailored for a target application. Each of the bent portions may be bent an amount up to 90°, for example. The bent portion **2600** may enable the first mixing plate **200** to provide a relatively low pressure drop, which may be advantageous in various applications.

(126) FIGS. **29-35** illustrate other embodiments where the first mixing plate **200** is bent along two axes to provide curved surfaces in multiple directions. In this embodiment, the axes are orthogonal, but other orientations of the axes, and additional axes, may be utilized such that the first mixing plate **200** is tailored for a target application.

(127) The first mixing plate **200** is bent a first angular amount K1 around the first axes and a second angular amount K2 about the second axes. In various embodiments, K1 is greater than or equal to R1 and K2 is greater than or equal to R1. For example, K1 may be greater than R1 while K2 is equal to R1. In another example, K1 may be greater than R1 while K2 is equal to 5R1. This arrangement of the first mixing plate **200** provides fine tuning of the pressure drop and tumble provided by the first mixing plate **200**. Additionally, this arrangement increases shear force across the first mixing plate **200** to mitigate formation of deposits on the first mixing plate.

(128) FIG. **36** illustrates another embodiment where the first mixing plate **200** and the second mixing plate **201** intersect one another. This intersection increases bulk swirl intensity and reduces the tumble flow of the exhaust to an area of the first mixing plate **200** before the intersection. The depth of the intersection, and therefore the overlap of the first mixing plate **200** and the second mixing plate **201**, is varied so as to tailor the mixer **121** for a target application. In various embodiments, the depth of the intersection is between 0.01R1 and 0.5R1, inclusive.

(129) FIGS. **37** and **38** illustrate another embodiment where the first mixing plate **200** and the second mixing plate **201** are rotated at angles about the central axis **205** of the tubular conduit **108** such that the mixer **121** is asymmetric when viewed from a downstream end or an upstream end. Such an arrangement may be beneficial when the inlet of the tubular conduit **108** receives asymmetrical flow. Additionally, such an arrangement may provide decreased backpressure, or provide desirable flow to the exhaust, which are desirable in various applications. FIGS. **39-45** illustrate a similar embodiment but additionally including the third mixing plate **301** and the fourth mixing plate **302**.

(130) FIGS. **40-42** illustrate one arrangement. As shown in FIG. **40**, the first mixing plate **200** is angularly separated from the second mixing plate **201** by 60 degrees. As shown in FIG. **41**, the third mixing plate **301** is angularly separated from the fourth mixing plate **302** by 180 degrees. A

view of the mixer **121** from an upstream end of the mixer **121** looking downstream is provided in FIG. **41**.

(131) FIGS. **43-46** illustrate another arrangement. As shown in FIG. **43**, the first mixing plate **200** is angularly separated from the second mixing plate **201** by 60 degrees. As shown in FIG. **44**, the third mixing plate **301** is angularly separated from the fourth mixing plate **302** by 60 degrees. However, in contrast to the arrangement of FIGS. **40-42**, the second set of mixing plates (the third mixing plate **301** and the fourth mixing plate **302**) is angularly separated (e.g., clocked) from the first set of mixing plates (the first mixing plate **200** and the second mixing plate **201**). A view of the mixer **121** from an upstream end of the mixer **121** looking downstream is provided in FIG. **45**, and a view of the mixer **121** from a downstream of the mixer **121** looking upstream is provided in FIG. **46**.

(132) In various embodiments, the first mixing plate **200** and the second mixing plate **201** are coupled to the tubular conduit **108** using tabs **4700** that extend through slots **4702** in the tubular conduit **108**, as shown in FIG. **47**. The tabs **4700** are positioned through the slots **4702** and are coupled (e.g., via welding, etc.) to the tubular conduit **108** (e.g., around the slots **4702**, etc.). In some embodiments, the tabs **4700** and slots **4702** are asymmetrical or otherwise configured for poke-yoke installation of the first mixing plate **200** and the second mixing plate **201**. The tabs **4700** are shown in FIG. **48**. In some embodiments, the tabs **4700** are folded over prior to coupling to the tubular conduit **108**.

(133) FIG. **49** illustrates the first mixing plate **200** according to various embodiments. In these embodiments, the first mixing plate **200** is aerofoil shaped. As a result, the first mixing plate **200** extends from a face that is oriented towards the flow of the exhaust to a tail that is oriented away from the flow of the exhaust. The exhaust flows along extended sloped surfaces between the face and the tail. A thickness of the first mixing plate **200** may be varied between the face and the tail. For example, the first mixing plate **200** may be thicker at the tail than the face, or may be thicker at the face than the tail.

(134) In various embodiments, as shown in FIGS. **50** and **51**, the first mixing plate **200** includes a plurality of notches **5000** (e.g., cutouts, etc.) in the first plate concave edge **210**. These notches **5000** provides higher localized shear stress that can reduce deposit formation along the first plate concave edge **210**. Additionally, the notches **5000** can be utilized to tailor the pressure drop provided by the first mixing plate **200**, and/or to cause the first mixing plate **200** to provide swirl. The shape, size, arrangement, and pattern (e.g., symmetric, asymmetric) of the notches **5000** may be varied such that the first mixing plate **200** is tailored for a target application.

(135) As shown in FIG. **52**, the first mixing plate **200** includes arcuate grooves **5200** on a face of the first mixing plate **200**, according to various embodiments. In some embodiments, each of the arcuate grooves **5200** extends around the first plate concave edge **210**, as shown in FIG. **52**. The arcuate grooves **5200** guide the exhaust across the first mixing plate **200** to encourage swirl and reduce intensity of tumbling motion by mitigating movement of the flow away from the face of the first mixing plate **200**.

(136) In various embodiments, the first mixing plate **200** includes a perforated portion that includes a plurality of perforations **5300**, as shown in FIG. **53**. The perforations **5300** may be uniformly distributed within the perforated portion, as shown in FIG. **53**. In some embodiments, the perforated portion is square. The perforations **5300** may also be distributed symmetrically or asymmetrically within the perforated portion. The perforations **5300** may be utilized to reduce backpressure of the first mixing plate **200**, or to target regions of a flow to deliver increased or decreased amounts of the treatment fluid.

(137) As shown in FIGS. **54-55**, the first mixing plate **200** and the second mixing plate **201** also include a plurality of connection cut-outs **5400**. Each of the connection cut-outs **5400** is configured to facilitate flow of the exhaust between the first mixing plate **200** and the tubular conduit **108** and between the second mixing plate **201** and the tubular conduit. The connection cut-outs **5400** may be

variously configured (e.g., symmetrical, asymmetrical, shape, size, etc.) such that the mixer **121** is tailored for a target application. The connection cut-outs **5400** may be utilized to reduce backpressure, or to target regions of a flow to deliver increased or decreased amounts of the treatment fluid. Additionally, the connection cut-outs **5400** provide increased wall shear along the tubular conduit **108** proximate the connection cut-outs **5400**. This increased wall shear may mitigate formation of deposits.

(138) In various embodiments, as shown in FIG. **56**, the first mixing plate **200** includes a plurality of protrusions **5600** (e.g., ribs, etc.) along the first plate concave edge **210**. Each of these protrusions **5600** provides a localized vortex generator for the exhaust.

IV. Configuration of Example Embodiments

(139) While this specification contains many specific implementation details, these should not be construed as limitations on the scope of what may be claimed but rather as descriptions of features specific to particular implementations. Certain features described in this specification in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can, in some cases, be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

(140) As utilized herein, the terms “substantially,” “generally,” “approximately,” and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the appended claims.

(141) The term “coupled” and the like, as used herein, mean the joining of two components directly or indirectly to one another. Such joining may be stationary (e.g., permanent) or moveable (e.g., removable or releasable). Such joining may be achieved with the two components or the two components and any additional intermediate components being integrally formed as a single unitary body with one another, with the two components, or with the two components and any additional intermediate components being attached to one another.

(142) The terms “fluidly coupled to” and the like, as used herein, mean the two components or objects have a pathway formed between the two components or objects in which a fluid, such as air, treatment fluid, an air-treatment fluid mixture, exhaust, hydrocarbon, an air-hydrocarbon mixture, may flow, either with or without intervening components or objects. Examples of fluid couplings or configurations for enabling fluid communication may include piping, channels, or any other suitable components for enabling the flow of a fluid from one component or object to another.

(143) It is important to note that the construction and arrangement of the various systems shown in the various example implementations is illustrative only and not restrictive in character. All changes and modifications that come within the spirit and/or scope of the described implementations are desired to be protected. It should be understood that some features may not be necessary, and implementations lacking the various features may be contemplated as within the scope of the disclosure, the scope being defined by the claims that follow. When the language “a portion” is used, the item can include a portion and/or the entire item unless specifically stated to the contrary.

(144) Also, the term “or” is used, in the context of a list of elements, in its inclusive sense (and not in its exclusive sense) so that when used to connect a list of elements, the term “or” means one,

some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, Z, X and Y, X and Z, Y and Z, or X, Y, and Z (e.g., any combination of X, Y, and Z). Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present, unless otherwise indicated.
(145) Additionally, the use of ranges of values (e.g., W1 to W2, etc.) herein are inclusive of their maximum values and minimum values (e.g., W1 to W2 includes W1 and includes W2, etc.), unless otherwise indicated. Furthermore, a range of values (e.g., W1 to W2, etc.) does not necessarily require the inclusion of intermediate values within the range of values (e.g., W1 to W2 can include only W1 and W2, etc.), unless otherwise indicated.

Claims

1. An exhaust aftertreatment assembly comprising: a tubular conduit including a central axis; and a mixer disposed in the tubular conduit, the mixer comprising: a crescent shaped first mixing plate including: a first plate convex edge attached to the tubular conduit, and a first plate concave edge that intersects the first plate convex edge at a first plate first point and a first plate second point; and a crescent shaped second mixing plate including: a second plate convex edge attached to the tubular conduit, and a second plate concave edge that intersects the second plate convex edge at a second plate first point and a second plate second point; wherein a first plane in which the first mixing plate extends and a second plane in which the second mixing plate extends are oblique to the central axis such that the first plane intersects the second plane inside of the tubular conduit.
2. The exhaust aftertreatment assembly of claim 1, wherein: the first mixing plate is flat; and the second mixing plate is flat.
3. The exhaust aftertreatment assembly of claim 1, wherein the first mixing plate and the second mixing plate are positioned relative to the tubular conduit such that the central axis extends between the first plate concave edge and the second plate concave edge.
4. The exhaust aftertreatment assembly of claim 1, wherein the first plane intersects the second mixing plate between the second plate first point and the second plate second point of the second plate convex edge.
5. The exhaust aftertreatment assembly of claim 1, wherein the first plate concave edge comprises: a plurality slots; and a plurality of tabs respectively positioned between adjacent slots of the plurality of slots, each tab extending away from a body of the first mixing plate.
6. The exhaust aftertreatment assembly of claim 5, wherein at least one first tab of the plurality of tabs extends over a side of the body of the first mixing plate, and at least one second tab of the plurality of tabs extends parallel to the side of the body.
7. The exhaust aftertreatment assembly of claim 1, wherein the mixer further comprises: a third mixing plate arranged downstream of the first and second mixing plates, the third mixing plate including: a third plate convex edge attached to the tubular conduit, wherein, in a view along the central axis, a midpoint of the third plate convex edge is offset 90 degrees about the central axis relative to a midpoint of the first plate convex edge.
8. The exhaust aftertreatment assembly of claim 1, wherein the mixer further comprises: a third mixing plate arranged downstream of the first and the second mixing plates, wherein, in a view along the central axis, the third mixing plate is offset 90 degrees about the central axis relative to the first mixing plate or the second mixing plate.
9. The exhaust aftertreatment assembly of claim 8, wherein the mixer further comprises: a fourth mixing plate positioned opposite the third mixing plate.
10. The exhaust aftertreatment assembly of claim 1, wherein the first mixing plate and the second mixing plate each include a first end portion, a second end portion, and a bent portion arranged

between the first end portion and the second end portion.

11. The exhaust aftertreatment assembly of claim 1, wherein the first mixing plate is bent a first angular amount about a first axis of the first mixing plate and a second angular amount about a second axis of the first mixing plate, the first angular amount being greater than the second angular amount.

12. The exhaust aftertreatment assembly of claim 1, wherein the mixer further comprises: a third mixing plate; a fourth mixing plate; a fifth mixing plate positioned between the first mixing plate and the second mixing plate; and a sixth mixing plate positioned between the third mixing plate and the fourth mixing plate.

13. The exhaust aftertreatment assembly of claim 1, wherein the mixer further comprises: a third mixing plate attached to the tubular conduit such that the first mixing plate, the second mixing plate, and the third mixing plate are offset from each other by 120 degrees about the central axis.

14. The exhaust aftertreatment assembly of claim 1, wherein the first mixing plate intersects the second mixing plate.

15. The exhaust aftertreatment assembly of claim 1, wherein the first mixing plate and the second mixing plate are positioned at angles about the central axis such that the mixer is asymmetric when viewed from an upstream or downstream end of the tubular conduit.

16. The exhaust aftertreatment assembly of claim 1, wherein the first mixing plate includes at least one of (a) a plurality of notches formed along the first plate concave edge (b) a plurality of arcuate grooves formed on a first face of the first mixing plate so as to extend around the first plate concave edge (c) a plurality of cut-outs formed along the first plate convex edge, or (d) a plurality of protrusions formed along the first plate concave edge.